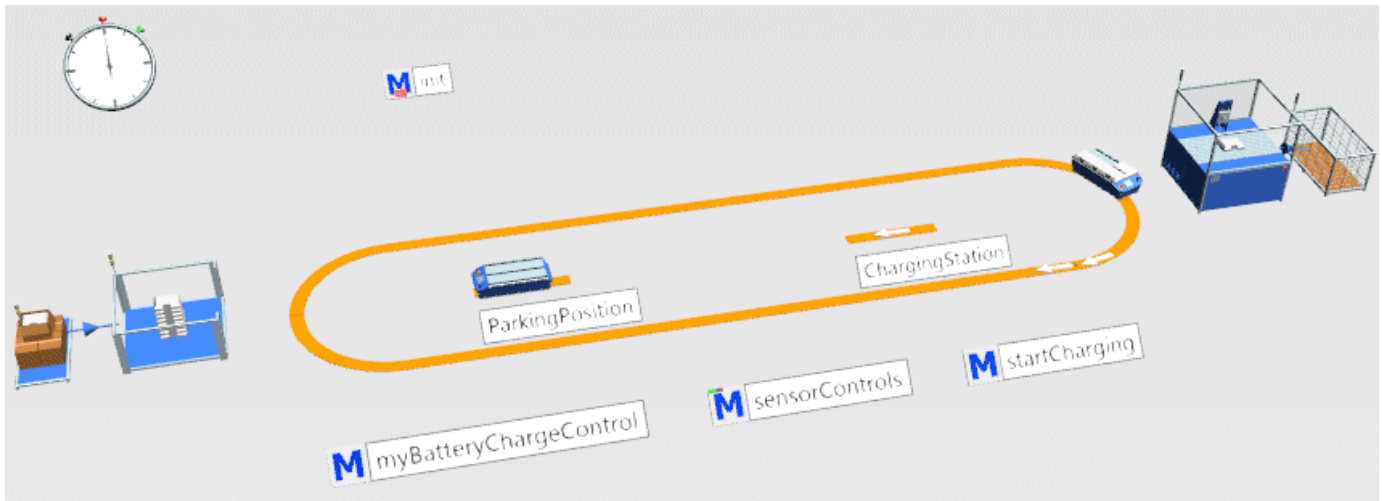




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Preparing Data for the Simulation with DataFit

Prepare Data for the Simulation with DataFit

One of the purposes of your simulation study will be to optimize the availability of resources, such as machines, laborers, vehicles, etc. for an existing or a planned production plant. This means that you have to reduce the number of failures and down-times of machines. When a machine fails, it has to be repaired, which leads to delays in the production and costs money.

Oftentimes the *Mean Time Between Failures* (MTBF) and the *Mean Time To Repair* (MTTR) are stochastically distributed. Because of the qualitative properties of a system with failures, we know the type of distribution for the MTBF and for the MTTR. What we do not know are the parameters of the distribution.

Production planning specialists do observe the availability of resources, the MTTR and the MTBF. The availability is the MTBF divided by the MTBF plus the MTTR times 100 percent, i.e.,

$$AV = \frac{100 \cdot MTBF}{MTBF + MTTR}$$
 So when we know two of the three values—AV (availability), MTTR and MTBF—we can compute the third one. The *DataFit Wizard* computes the parameters of the distributions for the MTTR and MTBF to arrive at the availability you want to achieve.

Note

We modeled the *DataFit objects* in *Frames* and with the object *Dialog*. For this reason they do not use the Siemens PLM standard design, but the Windows design.

Compare [Statistics Tools](#)

Go to [Step 1: Define the Task and the Objectives](#)

Go to [Step 2: Collect and Prepare Data](#)

Go to [Step 3: Decide Which Distribution to Use](#)

Step 1: Define the Task and the Objectives

You will start out with defining the task and the objective of the simulation study, being careful to not predefine any one solution you might expect or want to be true.

Some of the questions you want answers for are:

- What throughput and what output can we expect?
- What is the optimal number of resources (machines, workers, tools)?
- Where in the plant are buffers necessary? What is the optimal buffer size?
- What is the optimal number of work piece carriers?
- Which control strategies are best suited for the task?
- How do some or all of the above factors interact and produce different results?

Then, you will have to decide the scope of the simulation.

- Do you only need to simulate the production plant?
- Or do you also have to simulate other areas, such as receiving the parts delivered by the suppliers, warehousing, shipping, etc.?

Go to [Step 2: Collect and Prepare Data](#)

Go to [Step 3: Decide Which Distribution to Use](#)

Step 2: Collect and Prepare Data

Collecting and preparing the data to use in the simulation study is one of your most important tasks.

This takes up about 35 percent of the project time! Creating the model takes up 25 percent, validating and correcting 15 percent, running experiments 10 percent, analyzing and evaluating 15 percent of the project time.

You will use:

Static data, such as:

- Data extracted from the production program, which you collected by watching the machines in the plant or which you received from the experts on the shop floor.
- Data extracted from the structure of the plant.
- Data from a shift model.

Stochastic data, such as:

- The failure behavior.
- The processing and set-up times.
- The amount of rejects.

For simulating the material flow of interlinked production and assembly facilities you need:

Procedure

1. Data covering the entire production system (**model data**):
 - The production program, such as the produced products, the lot sizes, etc.
 - The layout of the plant, its structure, controls, and the points where parts are introduced into or discharged from the system.
 - Reworking, such as reworking times, the station that produces rejects, and the frequency of rejects.
2. Data covering the staff working in the plant (**object data**):
 - The number of workers, and their qualifications.
 - The shifts worked.
 - The distances covered and the required times.
3. Data covering machines, transport systems and work stations (**object data**):
 - Failures, including MTBF, MTTR, reference time, distribution, and variance.
 - Cycle times, capacities and cycle variance.
 - Set-up times.
 - Supply with materials.

Back to [Step 1: Define the Task and the Objectives](#)

Go to [Step 3: Decide Which Distribution to Use](#)

Step 3: Decide Which Distribution to Use

Step 3: Decide Which Distribution to Use

Once you have collected the data you need for the simulation study, you have make a number of decisions.

- Which distributions to select for the **Processing Time**, the **Set-up Time**, the **Recovery Time**, the **Cycle Time**, and the **Failure Times** for the *Plant Simulation* objects.
- Which parameters to enter for the selected distribution.

For this, we provide *Statistics Wizards*.

You can:

- [Practice Distribution-Fitting with DataFit.](#)
- [Use Distributions with Bounds.](#)

Compare [Model Random Processes](#)

Back to [Step 1: Define the Task and the Objectives](#)

Back to [Step 2: Collect and Prepare Data](#)

Distribution-Fitting with DataFit

Use the object *DataFit* for distributions without bounds.

Proceed as follows:

- Enter a sample of observations of a random number and a **Level of significance**.
- Then, the *Wizard* finds the parameters of the selected distribution.

The *Wizard* estimates the parameters of the selected distribution(s) and then performs a goodness-of-fit test with the filtered data on the tab **Fit**.

Note

The **Level of significance** is the probability with which a distribution with estimated parameters will be rejected by the goodness-of-fit test, although it is suitable for the sample.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.

Compare [Statistics Tools](#)

Back to [Step 3: Decide Which Distribution to Use](#)

Input Data in DataFit

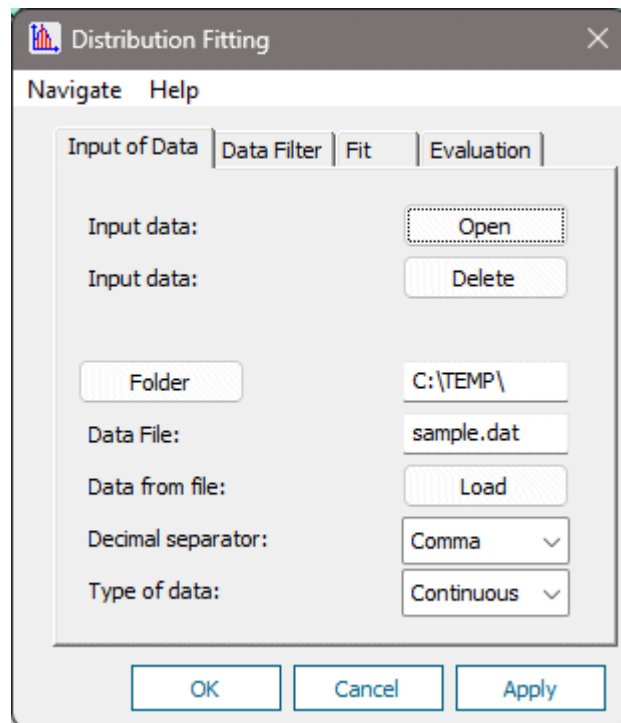
In this step we input data into *DataFit*.

Note

DataFit only computes the distribution parameters for a sample with more than 10 values.

You can only proceed to the next tab, after you have selected the required settings and typed the data for them in on the active tab. *DataFit* then activates buttons and/or text boxes on the next tab.

Most buttons are toggle buttons that turn a feature on or off.



Import data to be fitted on the tab **Input**

- Import the data you received—sample data observed on the shop floor, or the results of a previous simulation run—into a column of a *Plant Simulation* table file. Then, drag this table over the object *DataFit* in the *Frame* that contains your simulation model and drop it there.
- Select a column that contains the data you want to use in the dialog that opens and click **Load**.
Or
- Click **Read** to import data from the file whose name you type into the text box **Data file**.
- When you import data from other programs, for example from Excel, you have to select the same **Decimal separator** in *Plant Simulation* is you used in the other application. You might also paste the sample data into the table via the Windows-Clipboard. For this you have to click **Open** to open the input table.
- Select which type of data you want to investigate:
 - **Discrete** numbers, i.e., any integer number greater than or equal to 0, this is 0, 1, 2, 3, etc.

- Continuous numbers, which include negative numbers, and floating point numbers, for example -10, 1.4, 5, ...

When you have typed in all required data or selected the required settings, you can proceed to the tab **Filter**.

Back to [Distribution-Fitting with DataFit](#)

Filter Data in DataFit

In this step we filter data in *DataFit*.

Filter the data on the tab **Filter**.

The screenshot shows the 'Data Filter' tab in the DataFit software. The interface includes a tab bar at the top with 'Input of Data', 'Data Filter' (selected), 'Fit', and 'Evaluation'. Below the tab bar, there is a checkbox labeled 'Filter data' which is checked. To the right of the checkbox are four input fields: 'Lower bound (all values >):' with the value '0', 'Upper bound (all values <):' with the value '1', 'Offset (all values +):' with the value '0', and 'Zoom factor (all values *):' with the value '1'. At the bottom of the tab, there are two buttons: 'Descriptive statistics:' with an 'Open' button next to it, and 'Sorted data:' with an 'Open' button next to it.

Filtering serves three purposes:

Procedure

1. It cuts off data that is atypical.
2. It eliminates outliers, i.e., it removes these values from the sample.
3. It transforms data to a certain range, i.e., it translates or zooms data.

To filter data:

- You can, but do not have to enter a value for the **Lower bound** and a value for the **Upper bound**.
- Click **Start** to start filtering the data.
- This activates the buttons **Show** descriptive statistics and **Open** filtered data.
 - Clicking **Show** opens a table with descriptive statistics of the characteristics of the sample data.

- Clicking **Open** filtered data opens a table with the sorted data to be used for data fitting, i.e., after eliminating unwanted values.

Once you have entered the required data or selected the required settings, you can proceed to the tab **Fit**.

Back to [Distribution-Fitting with DataFit](#)

Fit Data in DataFit

In this step we fit data in *DataFit*.

DataFit estimates the parameters of the selected distribution(s) and then performs a goodness-of-fit test with the filtered data on the tab **FIT**.

- Click **Select** and select one, several or all distributions in the dialog that opens.
- Enter a **Level of significance** for the values. You can enter a number between 0 and 1. In most cases you will enter a number between 0 and 0.2.
- Click **Fit** to estimate the parameters for the selected distribution or distributions using the sample.
- If you are investigating a large sample, you will have to use the *Chi-Square Test* is the Goodness-of-Fit Test. For this you have to enter the **Number of classes**. The Goodness-of-Fit Test describes how good the distribution with its parameters fits the given sample.
- Click **Show** to display a histogram of the sample in a chart.

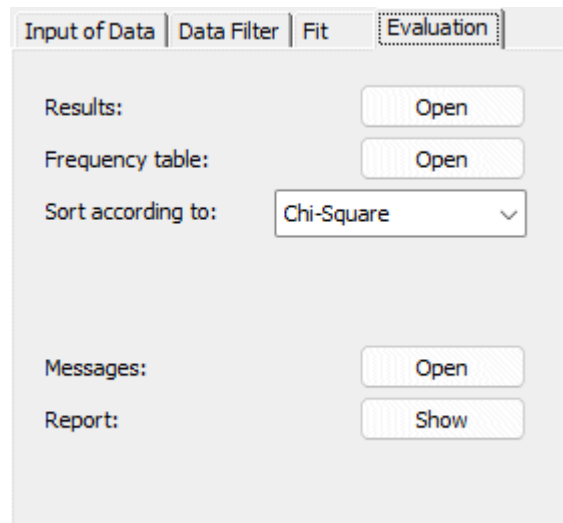
Once you have entered the required data or selected the required settings, you can proceed to the tab **Evaluation**.

Back to [Distribution-Fitting with DataFit](#)

Evaluate Data in DataFit

In this step we evaluate data in *DataFit*.

Perform an analysis of the filtered data on the tab **Evaluation**.



Select how *DataFit* sorts the fitting tests:

- The **Chi-Square** test suits all purposes. It splits up the sampled data into classes. The sample has to have more than 40 values.
- The **Kolmogorov-Smirnov** and the **Anderson-Darling** test are only suited for continuous distributions with 10 to 2000 values.

Open the results table that shows which parameters of which distribution to enter in *Plant Simulation*.

- The columns **Result Chi**, **Result KS**, **Result AD** tell, if the distributions are applicable to the sampled values or not. This is the case, when the respective value statistic is less than the respective **value**. Naturally, the evaluation takes the **Level of significance** into account.
- The columns labeled **Parameter 1**, **Parameter 2**, and **Parameter 3** show the exact numbers and the names of the rounded parameters as you enter them into the dialogs of the *Plant Simulation* objects.
- To show a **Report** of the parameters and the goodness-of-fit test, click **Show**.
- To reduce the size of the model, which might have been bloated by the **Report**, click **Delete**.
- When none of the selected distributions adequately represent the properties of the sample, you can use an empirical distribution to model the data of the sample in *Plant Simulation*. Click **Open** to open the **Frequency table**.

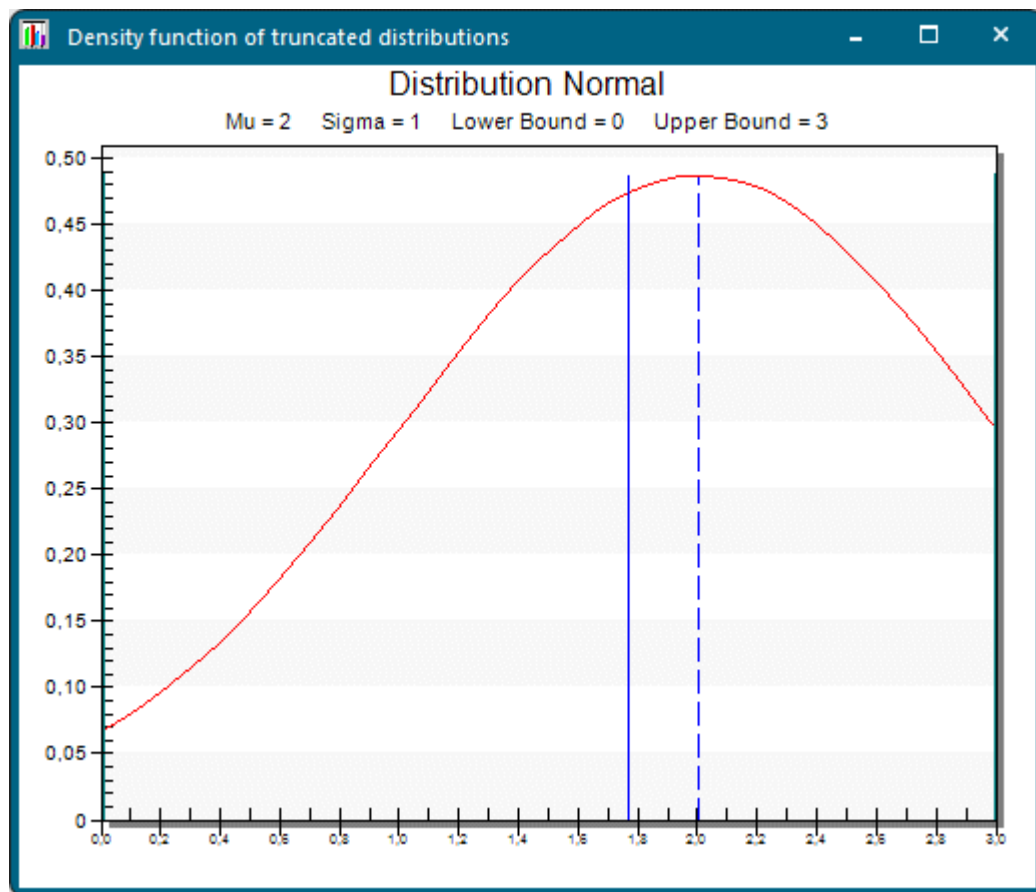
Back to [Distribution-Fitting with DataFit](#)

Use Distributions with Bounds

The distributions you use in your simulation have realizations that may be arbitrarily large or small, respectively. For this reason, you will use distributions with upper and lower bounds, to cut off values you do not want to consider.

Be aware that using a distribution with bounds will change the mean value and the standard deviation.

When you use the normal distribution for modeling a random time, the lower bound automatically will be 0.



When you type in the parameters $\mu = 2.0$, and $\sigma = 1$ for the normal distribution, a real mean value of the distribution with the lower bound 0 and the upper bound 3 will be 1.77.

Use the object *TruncDistribution*  for distributions with bounds.

- Select a distribution and both bounds.